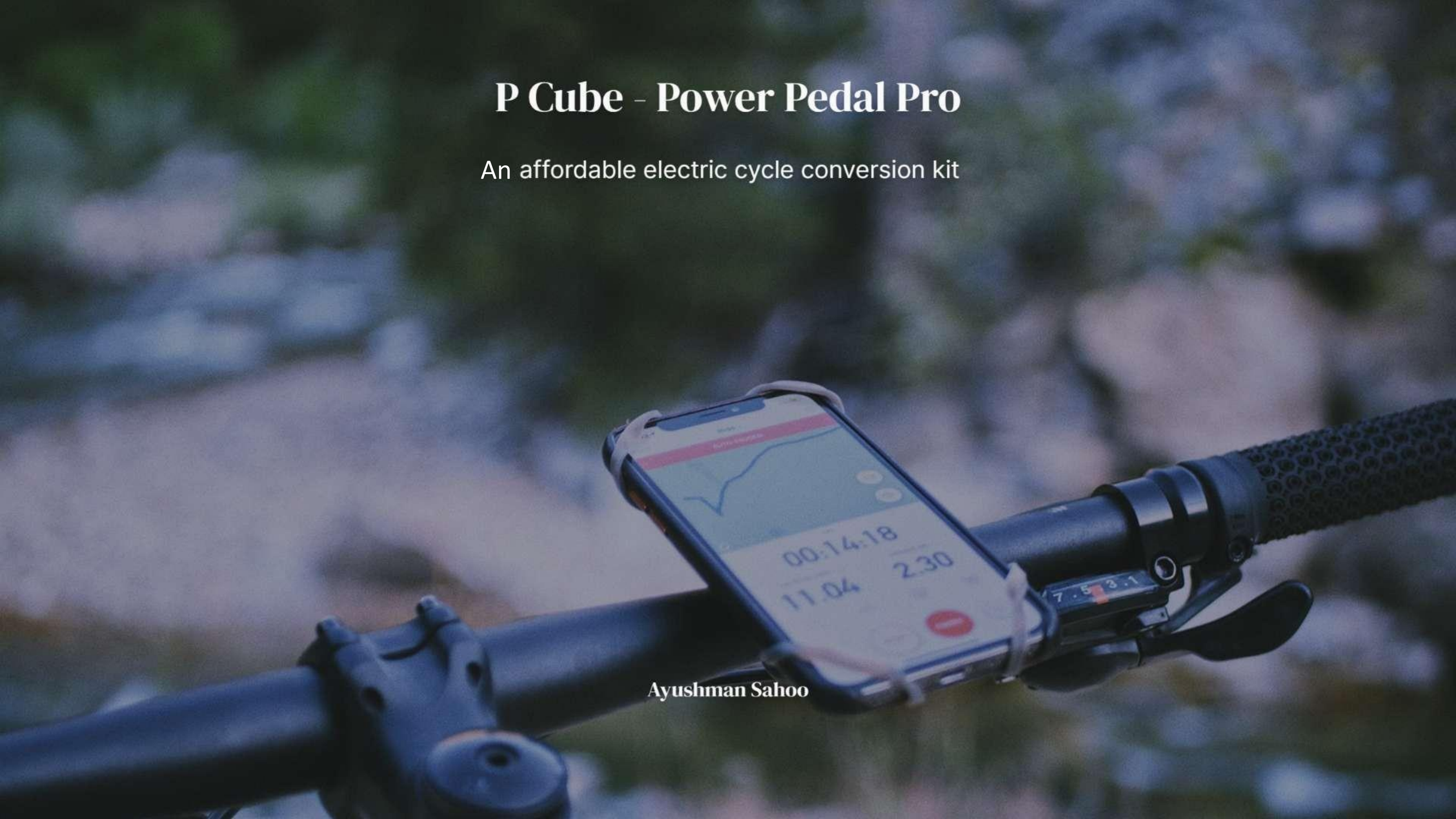
P Cube - Power Pedal Pro

An affordable electric cycle conversion kit



Ayushman Sahoo

Proof of Concept (PoC)

Demonstrating Feasibility and Practicality of an Affordable Electric Bicycle Conversion Kit

Objective:

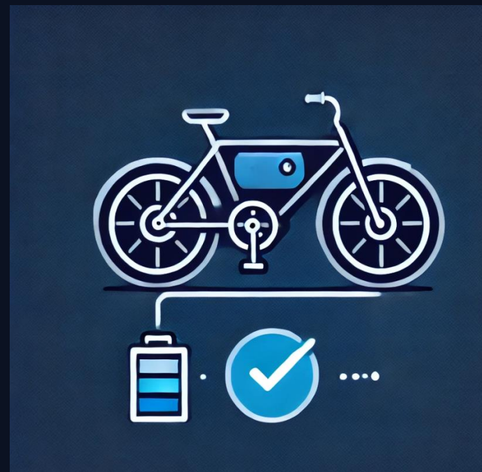
To validate the functionality and practicality of the P^3 kit as an affordable electric bicycle conversion option for low-income individuals.

Key Components Tested:

- **Performance:** Provides sufficient pedal-assist and power output for short to medium commutes.
- **Compatibility**: It can be easily installed on various types of bicycles.
- **Sustainability**: Uses recycled materials and sealed lead-acid batteries as an eco-friendly solution.

Initial Testing Results:

- **Efficiency**: Cam cover 15km in 5hr charge with max speed of 45km/h
- **User Feedback**: Been using the product for 1.5 years without major issues. Saves time and physical effort.



Problem Statement

Incomplete Conversion kits

Most conversion kits are often
Incomplete or too expensive

Expensive transportation

People face difficulties commuting
due to expensive transportation options

Need for reliable kits

There is a need for a low-cost, reliable solution
to make electric commuting accessible to all

Opportunities

Tapping into the Demand for Affordable, Eco-Friendly Transportation

Demand:

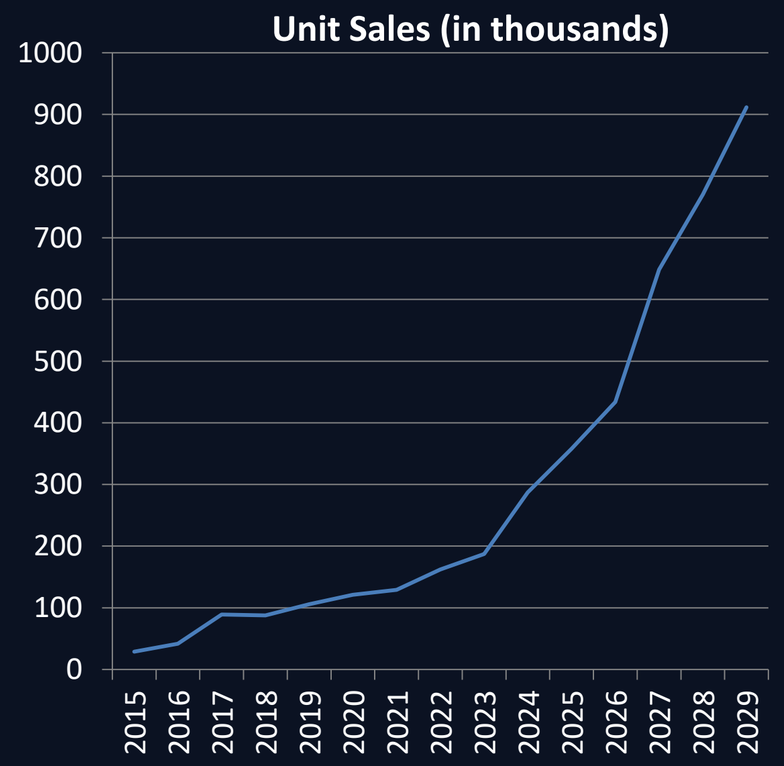
There is increase in demand for affordable, sustainable commuting solutions among urban populations. The graph shows the same.

Competitive Advantage:

P^3 is very affordable and is customizable without loosing functionality. It can be repaired by anyone due to its simplicity.

Public awareness:

Even poor people are looking towards cheaper alternatives to motorcycles. They want faster ways to commute.



Introduction to P Cube-Power Pedal Pro

A Revolutionary Electric Powered Cycle

Environmentally Friendly

With zero emissions and a pedal-assist feature, P Cube-Power Pedal Pro is environmentally friendly.

Sleek and Stylish Design

P Cube-Power Pedal Pro features a sleek and stylish design that is sure to turn heads.

Affordable and Efficient

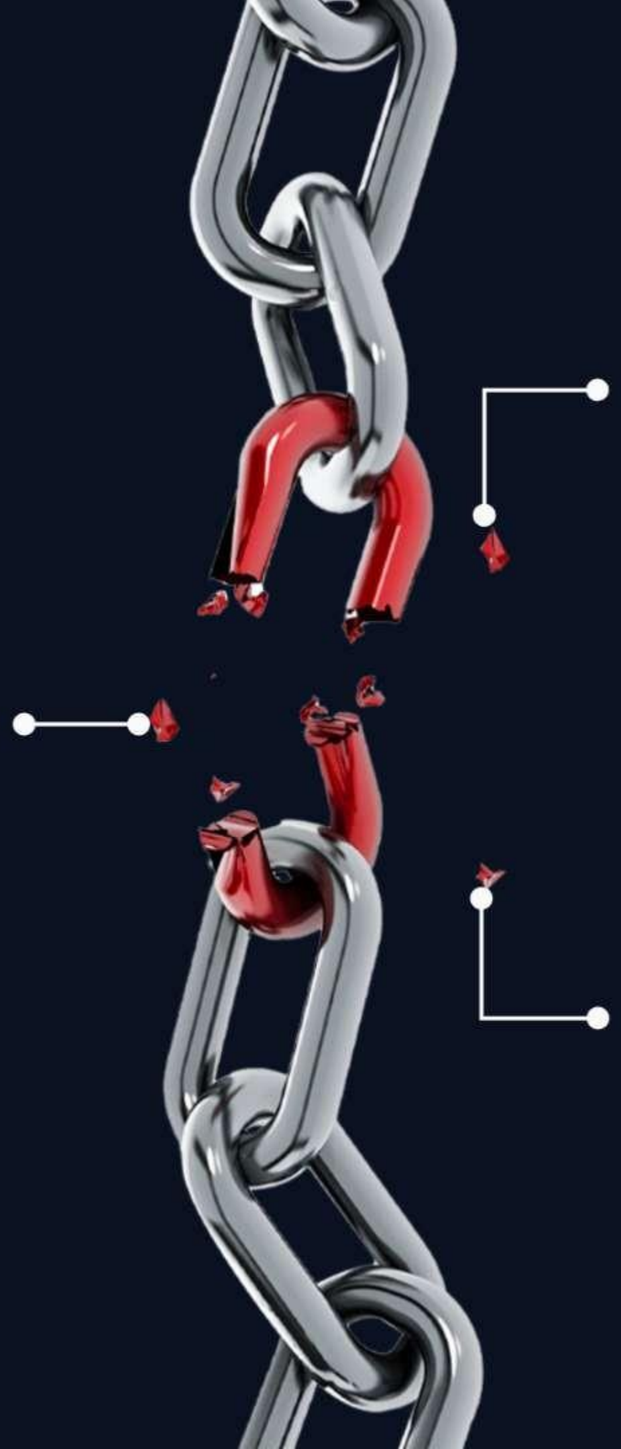
P Cube-Power Pedal Pro offers an affordable and efficient electric powered cycle solution.

How P Cube-Power Pedal Pro works

Unveiling the Technology Behind

Pedal-Assist Technology

P Cube-Power Pedal Pro uses pedal-assist technology to provide electric power while cycling.



Intelligent Motor Control

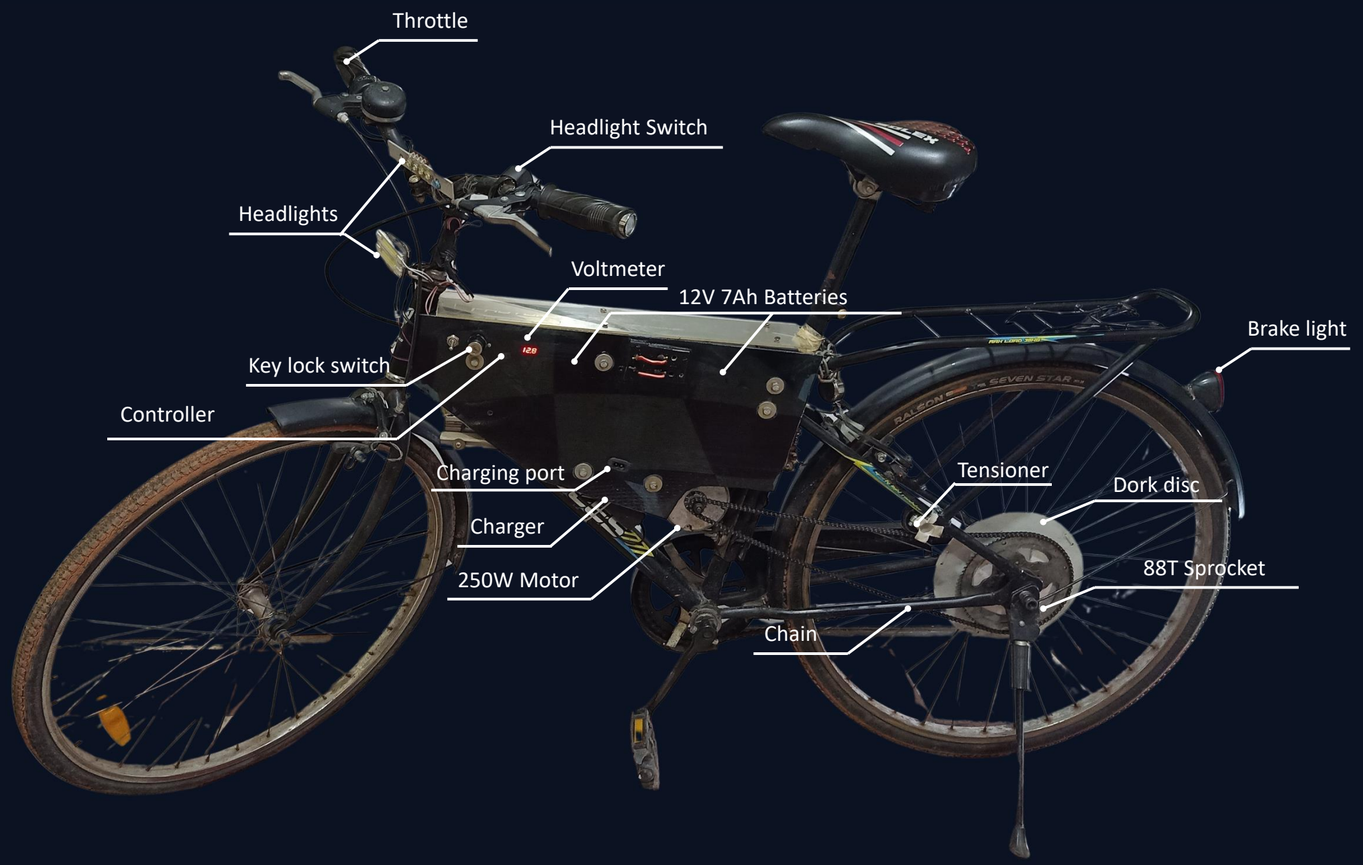
The intelligent motor control system ensures seamless integration of electric power with manual pedaling.

Battery Charging

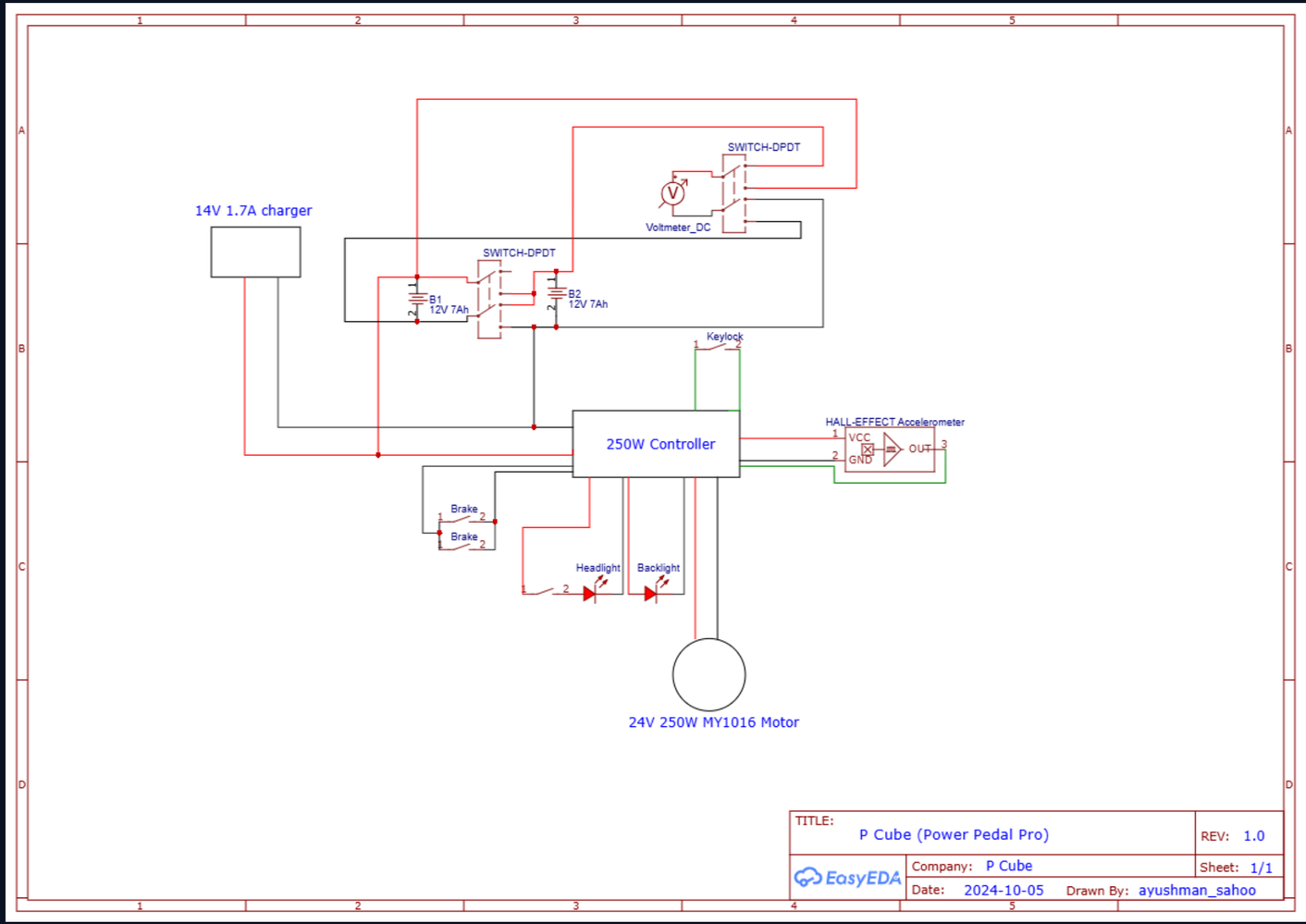
P Cube-Power Pedal Pro's battery can be easily charged using a standard power outlet.

Technical Overview

Exploring the technology that powers the P³



Circuit

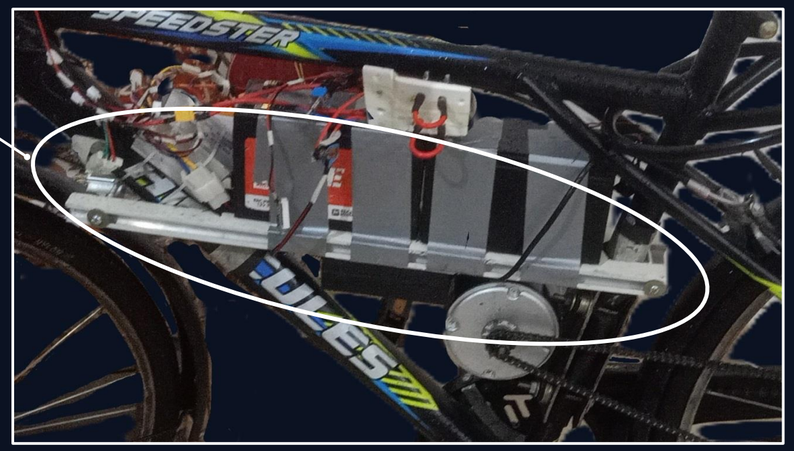

14V 1.7A charger
SWITCH-DPDT
Voltmeter_DC
SWITCH-DPDT
B1
12V 7Ah
B2
12V 7Ah
Keylock
250W Controller
HALL-EFFECT Accelerometer
VCC
GND
OUT
Brake
Brake
Headlight
Backlight
24V 250W MY1016 Motor
TITLE:
P Cube (Power Pedal Pro)
REV: 1.0
Company: P Cube
Sheet: 1/1
Date: 2024-10-05
Drawn By: ayushman_sahoo
EasyEDA

Technical Advantage

Engineering Advantages of Power Pedal Pro

Lightweight and Durable Aluminum Extrusion for Battery Bracket

- **Compatibility:** Can be fitted at any angle in any cycle frame
- **Strength and Durability:** Aluminum extrusion provides a strong yet lightweight framework, ensuring the battery is securely held while reducing the overall weight of the
- **Cost-Effective and Easy to Manufacture:** The extrusion process allows for precise, custom shapes, enhancing both cost efficiency and ease of assembly.
- **Sustainability**: Aluminum is a recyclable material, contributing to the overall eco-friendliness of the P Cube design



Use of recycled sealed lead acid

- **Proven Reliability:** Sealed lead-acid batteries are known for their robust performance and long lifespan, providing consistent power output for the system.
- **Cost-Effective and Accessible**: Recycled SLA batteries are more affordable than newer battery technologies, making the P Cube solution more accessible for low-income users without compromising on quality.
- **Safe and Maintenance-Free:** SLA batteries are sealed, preventing leaks and reducing the need for maintenance, which is crucial for users in resource-limited settings. Can be repaired by anyone without much expertise.

Preliminary Studies

Initial Studies and Prototyping Insights

Material Selection & Testing :

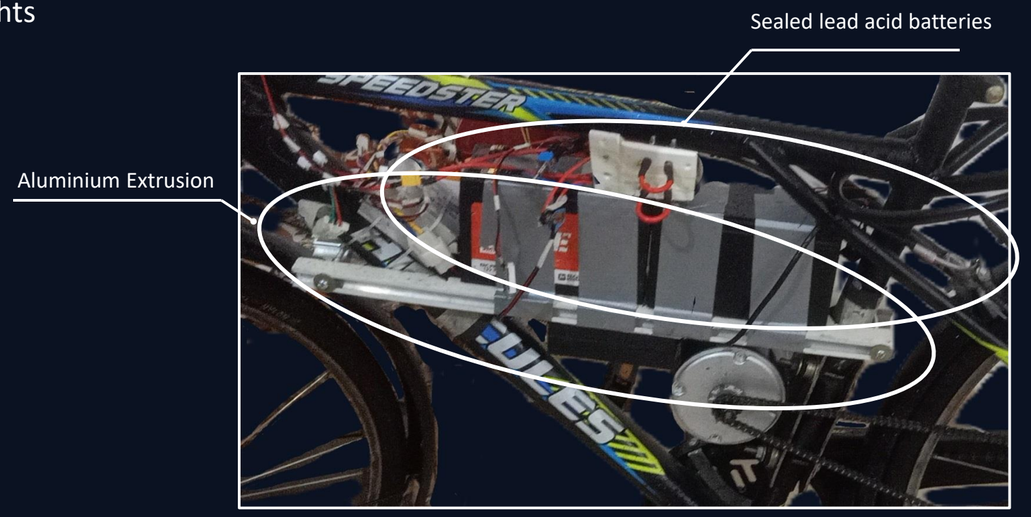
- **Aluminum Extrusion :** Tested for durability, strength, and weight efficiency. Ideal for the structural integrity of the system.
- **Recycled Sealed Lead-Acid Battery:** Evaluated for sustainability, availability, and cost-effectiveness. Aligns with project goals of affordability and sustainability.

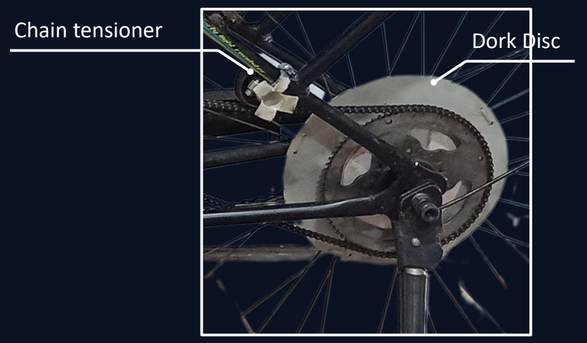
Redundancy testing:

- **Dork disc:** Initially made from laminated cardboard to prevent damage to the spokes if chain derails. Now is made from PVC sheet.
- **Chain tensioner:** There are many ways to maintain tension in the chain, but most effective was reused ball bearing from bike.

Motor selection :

- **Selecting the right motor was a difficult task. Brushless DC motors were too much to control. The final option was Brushed DC motors.**





Preliminary Studies

Initial Studies and Prototyping Insights

MY1016 350W:

- **Price: ₹ 2,199**
- **Torque: 1.04 N m**
- **Rpm: 2650**

MY1016Z 250W:

- **Price: ₹ 4,528 /-**
- **Torque: 8.15 N m**
- **Rpm: 330**

MY1016 250W:

- **Price: ₹ 1,922**
- **Torque: 0.8 N m**
- **Rpm: 2750**

Too much rpm but low torque. There was a need to increase torque.

11 Teeth

88 Teeth

Input gear: 11
Input RPM: 2650
Input torque: 0.8 N m

Output gear: 88
Output RPM: 343
Output torque: 6.4 N m

Power analysis:

- Battery voltage: 12 x 2 V
- Battery capacity: 336 W h
- Battery charge time: 4~5 hr

Motor voltage: 24 V
Motor current: 15 A
Estimated runtime: ~80 mins

Benefits of using P Cube-Power Pedal Pro

Enhance Your Cycling Experience

1

Effortless Commuting

Experience effortless commuting with the electric power-assist feature of P Cube-Power Pedal Pro.

2

Improved Fitness

P Cube-Power Pedal Pro combines the benefits of electric power and traditional cycling to improve fitness levels.

3

Cost Savings

Save on fuel costs and reduce your carbon footprint with P Cube-Power Pedal Pro.

4

Support Recycling

P Cube uses recycled materials to save environment and reduce cost

Competitive landscape and Value proposition

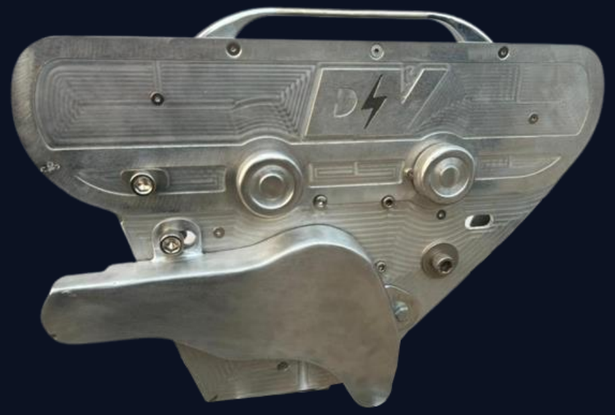
Why Choose P Cube-Power Pedal Pro?

Too expensive !!

₹38,890/-

Lacks basics like battery, charger, case

₹7,999/-



Made only for triangular frame

₹13,999/-

Completeness:

- P Cube provides complete components unlike other competitors. It includes all the things one would need,

Affordability:

- It targets low income people, hence, is cheaper than others.

Eco-Friendly Without Compromises:

- We use recycled materials from the best dealers to ensure quality while keeping the cost low.

Necessary minimum expenses

SL. No.	Name	Specifications	Price	Link	Offline	Ordered	Final
1	DC Motor	250W 24V	1711	DC MOTOR		YES	1922
2	Battery	24V 12Ah	1000	Battery	1000	YES	1000
3	Speed controller	24V 33A	569	ESC		YES	569
4	Charger	14V 2A	350	Charger	350	YES	350
5	Sprocket	80T OD(165) ID(54)	335	Sprocket		YES	335
6	Freewheel	22T OD(93) ID(34)	199	Freewheel	100	YES	180
7	Chain	25H	400	Chain	0	YES	0
8	Keylock switch	12V 1A	167	Keylock switch		YES	167
9	Wire	14 AWG	99	Wire		YES	99
10	Wire	22 AWG	58	Wire		YES	58
11	Throtle	23mm 1.56M	249	Throtle		YES	249
12	Double sided threaded hub	34mm	550	Hub	150	YES	150
		Total Price	5687				

Additional expenses

SL. No.	Name	Specifications	Price	Link	Offline	Ordered	Final
1	USB Converter	12V to 5V 3A	68	USB	Free	YES	0
2	Headlight	12V 1.2A	144	Light	20	YES	20
3	Voltmeter and ammeter	0-100V 50A	306	Voltmeter	180	YES	
4	Tactile switch	12mm x 12mm x 6mm	10	Tactile	20	YES	20
5	Heat shrink	1mm 5m	25	Heat shrink		YES	25
6	Heat shrink	2mm 3m	25	Heat shrink		YES	25
7	Solder wire	1mm 50g	140	Solder	140	YES	
8	Voltmeter	DC 0 - 100V	130	Voltmeter	130+180	YES	230
9	DPDT Switch	10A 220V On-Off-On + 6A On-Off-On	69	Switch	30+40	YES	70
10	Foam gasket	10mm x 2mm x 10m	110	Foam		YES	110
11	Motor and battery socket (MALE) x 2	XT30-U	58	Male		YES	58
12	Motor and battery socket (FEMALE) x 2	XT30-U	58	Female		YES	58
13	Connectors x 10	2 pins JST XH 2.54mm Pitch	70	Connector		YES	70
14	Charger connector	2.1mm, 5.5mm	29	Charge		YES	29
15	Push button switch	Self lock	99	Power switch	29	YES	29
16	Acrylic sheet	4' x 2' + 2' x 3"	800	Sheet	400	YES	400
17	Nut and bolt and warsher	M6 2"	120	Nut	85	YES	85
18	Motor plate	14cm x 8cm	N/A	N/A	70	YES	70
19	Chain	1.5 feet	150	Chain	60	YES	60
20	Heat shrink	5mm 1m	40	Heat shrink	15	YES	15
21	Lock	5 link	100	Lock	58	YES	50
22	VTR Switch	6 pin on/off	9	Vtr	10	YES	20
23	Philips connector and plug	cable with AC connector	50	Philips	15	YES	15
24	Balls	for axle	2 × 18	Balls	1 × 18	YES	18
25	Spokes	for 26"	120	Spokes	60	YES	60
26	Nuts and bolts and warshers	4 inch × 12	NA	Screw	250	YES	250
27	Battery plate	1.5 ft 1 inch	300	Na	100	YES	100
						YES	
		Discount					-37
		Delivery charges					137

Making charges

SL No.	Name	Specifications	Price
1	Lathe	Sprocket lathe and freewheel weld with sprocket	100
2	Drill	Motor plate holes	70
3	Welding	Sprocket welded to hub	10
4	Chain joining	Chain joined and wheel alignment	100
5	Lacing	Spokes joined to the wheel	150
6	Tube	Valve tube replaced	15
		Total Price 3030	7511

Estimated Pricing

Breakdown of costs

For Production of KIT-

Component and manufacturing costs : ~ 5000/-

On road price : ~ 7200/-

Profit per unit : ~ 2200/-

For Production of pre installed kit with cycle-

Component and manufacturing costs : ~ 5000/- + ~4000/- (for cycle)

On road price : ~ 10000/-

Profit per unit : ~ 1000/-

Target Audience and Market Potential

Potential Users & Market Opportunity

Low-Income Individuals and Families

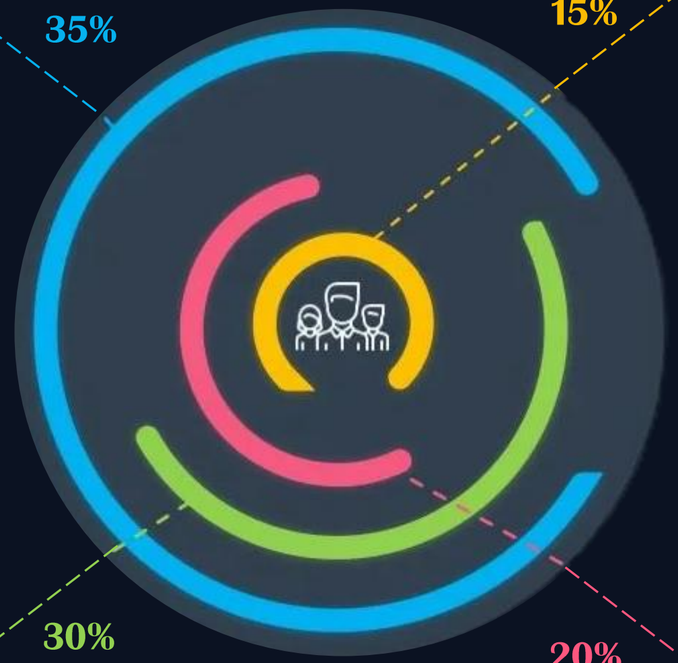
35%

Individuals with limited funds seeking affordable, reliable daily commuting options.

Urban Commuters

15%

Urban commuters seeking affordable, eco-friendly alternatives to motorbikes and public transport.



Daily Wage Workers and Laborers

30%

Workers needing affordable options for regular short to moderate distance travel.

Students and Young Adults

20%

Young commuters using bicycles for school, college, or work, seeking an affordable upgrade for easier travel.

Future Development and Scalability

Future Innovations & Growth Strategies

Adaptable Technology

The PPP kit is versatile and adaptable to various bicycle types, catering to a broad user base.

Enhanced Battery Technology

Research into using alternative, affordable battery types with longer life and faster charging.

Production Scaling

Can easily mass produced due to its simplicity. Increased production with same quality.

Additional Features

Exploring upgrades such as regenerative braking, app-based monitoring, and improved weather resistance.

Regional Expansion

Ability to expand the PPP kit to other regions within India with similar transportation challenges.

Collaborative initiatives

Collaboration with governments and NGOs to promote the P^3 through subsidies, enhancing access to affordable electric commuting.

Future Roadmap

Roadmap to Scale and Innovate

Phase 1: Product Refinement (0-6 months):

- **User Feedback Integration:** Collect and analyze feedback from initial users to make improvements in design, usability, and efficiency.
- **Prototype Optimization:** Further refine the prototype for enhanced performance and cost-effectiveness.
- **Material Sourcing:** Establish partnerships for consistent access to recycled materials.

Phase 2: Scaling and Mass Production (6-12 months):

- **Manufacturing Partnerships:** Secure partnerships for scalable, eco-friendly manufacturing to reduce costs while maintaining quality.
- **Full-Scale Deployment:** Launch the product on a larger scale in target regions.
- **Supply Chain Optimization:** Ensure smooth logistics and material sourcing to support increased production.

Phase 3: Market Expansion and Innovation (12-18 months):

- **Explore New Markets:** Expand the product offering into additional regions and explore new markets, including developed regions looking for sustainable energy solutions.
- **Technological Enhancements:** Introduce new features or updates based on advancements in technology and user needs.
- **Partnerships and Licensing:** Forge strategic partnerships with NGOs, governments, and private enterprises to increase reach and impact, while exploring licensing opportunities.

Phase 4: Long-Term Sustainability and Growth (18+ months):

- **R&D for Next-Gen Products:** Continue research and development to evolve the P Cube product line with more advanced, sustainable solutions.
- **Global Impact Initiatives:** Partner with global organizations to scale the solution worldwide, contributing to energy sustainability and climate action.
- **Community Empowerment Programs:** Develop educational programs to teach users how to maintain and maximize the use of the product, creating long-term self-sufficiency.

Conclusion and recommendation for P Cube-Power Pedal Pro

Make the Switch to Electric Powered Cycling

Experience the Future of Cycling

P Cube-Power Pedal Pro offers a glimpse into the future of cycling with its electric power-assist technology.

Join the P Cube-Power Pedal Pro Community

Become part of a growing community of electric powered cycling enthusiasts with P Cube-Power Pedal Pro.

Upgrade Your Cycling Experience

Upgrade your cycling experience with P Cube-Power Pedal Pro and enjoy the benefits of electric power.

Experience the Power of P Cube-Power Pedal Pro

Unleash the potential of your cycling journey with our affordable and efficient electric powered cycle

THANK YOU